

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Semester Midterm
Duration: 1 hour 20 min

Semester: Spring 2025
Full Marks: 40

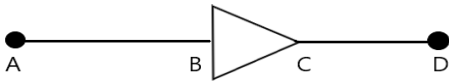
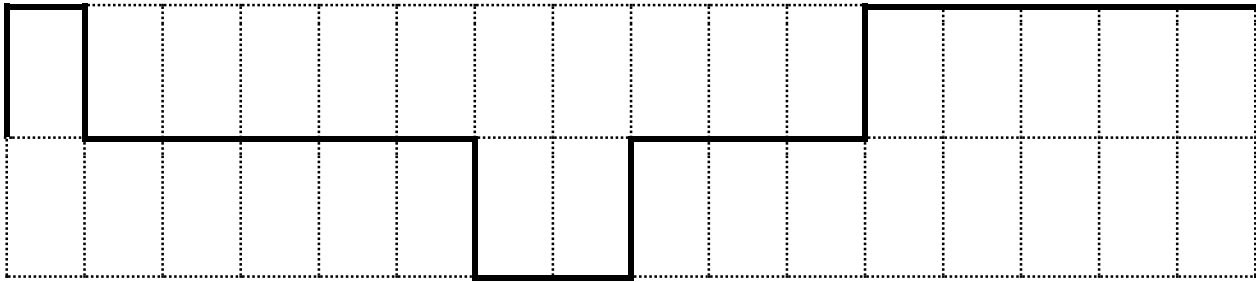
CSE 320: Data Communications

Answer the following questions [Total 3].
Figures in the right margin indicate marks.

SET A

Name:	ID:	Section:
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1.	a)	[CO1] Design a network infrastructure for a University Library that consists of three main units: i) Library Hall – This unit contains four subunits: Reading Area, Printing & Scanning Desk, Digital Resource Center and Circulation Desk. Each subunit is directly connected to a central device, ensuring high-speed communication and easy troubleshooting. ii) Book Storage & Archive Section – This unit consists of three subunits: Physical Book Storage, Digital Archive Server and Restricted Access Section. In this unit, all devices are interconnected, providing multiple communication paths to enhance reliability and prevent single points of failure. iii) Library Staff Office – This unit manages library operations and consists of four subunits: Librarian’s Office, IT & Technical Support Desk, Meeting Room and Administration & HR Office. Here, data flows in a structured sequence, with each device connected to two others, forming a continuous communication loop that prevents network collisions. The three main units are connected in a way that allows communication through a single backbone cable. Identify the networking structure used in each unit based on the characteristics described above. Draw a hybrid network topology diagram representing this setup, labeling all units and subunits accordingly. Calculate the total number of connection links.	[6]
	b)	[CO1] Highlight the key functionalities of the Transport layer and Internet layer of TCP/IP protocol stack. Mention which addresses are used here and why using both is necessary?	[4]

2.	a)	 [CO2] Suppose, a signal is transmitted from source A to destination D. An amplifier is placed between B and C. Power of the signal at source A is 20W. Attenuation along path A to B is 10 dB, and along path C to D is 1.76 dB. If power at destination is the same as source, find out how much the amplifier amplifies the signal.	[5]
	b)	[CO2] Suppose, while playing League of Legends, you are downloading a frame from a server that is 12,000 km away. The fiber-optic transmission medium from the server to your house contains 10 routers, each of which has a queuing time of 2 ms and a processing time of 1 ms. The speed of light inside fiber optic cable is 2×10^8 m/s, and the bandwidth of your network is 1 Mbps. If the latency (total delay) is found to be 0.098 s, calculate the size of the frame.	[5]
	c)	[CO2] Consider a transmission channel in which the value of SNR is zero. What can be deduced from this statement? A 1080p YouTube video has an average size of 25 MB (mega-byte) per minute. Suppose, you want to upload a 10-minutes long video. However, your network has a noisy channel with an SNR of 127. Consider the bandwidth to be 2.5 MHz. Calculate the maximum capacity bitrate of your network.	[5]
3.	a)	[CO2] Sami and his friend Shafi come across a three-level signal and both attempt to decode it. Sami claims that the signal follows the Bipolar AMI encoding scheme, while Shafi argues that it is encoded using MLT-3. Determine which of them is correct and then decode the given signal. [answer in question paper] 	[4]
	b)	[CO2] Deduce the appropriate scrambling method for the signal given in 3. a) and redraw the signal using it. [answer in question paper] 	[4]

c)	[CO2] Design a 3B/4B Block Coding scheme while following the conditions below. For convenience, the 4B encoded sequence for some 3B data sequences are already given. [Any valid combination following the conditions are correct] • The encoded sequence cannot contain more than two 0 • The encoded sequence cannot contain three or more consecutive 1 <table><tr><th>Data Sequence</th><th>Encoded Sequence</th></tr><tr><td>000</td><td>1101</td></tr><tr><td>001</td><td></td></tr><tr><td>010</td><td>1011</td></tr><tr><td>011</td><td></td></tr><tr><td>100</td><td></td></tr><tr><td>101</td><td>0011</td></tr><tr><td>110</td><td></td></tr><tr><td>111</td><td></td></tr></table> Based on the above block coding scheme, encode the following dataword into its corresponding codeword. 100010111001101	Data Sequence	Encoded Sequence	000	1101	001		010	1011	011		100		101	0011	110		111		[7]
Data Sequence	Encoded Sequence																			
000	1101																			
001																				
010	1011																			
011																				
100																				
101	0011																			
110																				
111																				

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